ABUSE-RING SENTINEL

End-to-End System Architecture for Abuse-Ring Detection & Risk Management

1 INGESTION LAYER

Accounts registry

Devices
Hardware fingerprints, IMEIs, canvas, cookies

Payments
Card hashed, bank accounts, IMPI VPA

Addresses
Delivery addresses, geo-coordinates

Ips
IP addresses, subnets, sessions

Orders
Transactions, amount, timestamp, category

Referrals
Referrer, referred, bonus credits

Normalized & Type-Casted Relational DataFrames first_seen_date tagged

6,558
6,591
9,601
9,681
6,594
18,966
2,240

2 GRAPH & RESOLUTION LAYER

1. ENTITY RESOLUTION
Collapse aliases, masks & noise to true physical entities

2. BIPARTITE PROJECTION
Inverse-frequency-weighted account-to-account edges: each shared resource contributes 1/degree (resource) to edge weight

3. LOUVAIN COMMUNITY DETECTION
Modularity Optimization

70 Candidate Syndicate Clusters
668 Accounts Monitored

3 DUAL FEATURE ENGINE

Engine A
Graph Topological Metrics

- cluster_size
- entity_reuse_ratio
- internal_density
- mean/max degree centrality
- mean/max PageRank
- mean/max betweenness
- creation_span_days
- creation_std_days
- avg_order_amount
- order_amount_cv

Engine B
Directed Referral Dynamics

- referral_cycle_ratio
- referral_resource_overlap_ratio
- median_referral_activation_days
- within_cluster_referral_density

cluster_features.csv
70 clusters x 17 engineered features

4 CALIBRATED ML & THRESHOLD LAYER

Champion Model Pipeline
StandardScaler + Logistic Regression on 7 champion features (3 graph + 4 referral)

5-Fold Stratified CV
Out-of-Fold Calibrated Probabilities

Pooled Cost-Optimal Threshold
15 seeds, 954 clusters, 292 true rings

0.1333
Locked Operating Threshold

10,000-Resample Bootstrap
Precision 1.000 [CI 1.000-1.000]
Recall 1.000 [CI 1.000-1.000]

5 TRIAGE COCKPIT & AUDIT EXPORT

Executive Scorecard

26 Flagged Clusters	Rs 7.08L Protected	~399x ROI

Interactive D3.js Physics Network

Single-Account Ego-Graph Bipartite-Tree Toggle

SHAP Waterfall (Top Features)

Hash-chained audit log
SHA-256, byte-reproducible audit_log.jsonl

Executive PDF Dossier Export
Cluster summary, network, features, SHAP, scores, audit trail & evidence pack – ready for review.

ZERO-LOOKAHEAD TEMPORAL BACKTEST

- **101** Point-in-Time Snapshots (7-Day Intervals)
- **28/28** Rings Detected (100%)
- **30.0-Day** Median Detection Latency
- **90.4%** Fraud Volume Intercepted Before Clearance
- **Rs 6.71L** Counterfactual Value Protected

DEFENSE-ONLY GUARANTEE

TECH STACK

Python, Pandas, NumPy, Faker, NetworkX, python-louvain, Scikit-Learn, D3.js, Streamlit, Shap, ReportLab, Vanilla CSS, Altair, Matplotlib, Docker

ABUSE-RING SENTINEL

Mathematical & Model Flow for Abuse-Ring Detection & Risk Management

1 BIPARTITE PROJECTION & EDGE WEIGHTING

$$w(u,v) = \Sigma\, 1 / \deg(r) \quad \text{for } r \in \text{shared\_resources}(u,v)$$

- Two accounts get an edge if they share ≥1 device, payment instrument, address, or IP.
- Each shared resource contributes 1/deg(r) to the edge weight — deg(r) is the number of accounts using that resource, so a resource used by 2 accounts contributes 0.5, one used by 20 accounts contributes 0.05.
- Resources unique to one account (deg<2) contribute nothing.

2 LOUVAIN COMMUNITY DETECTION

$$Q = (1/2m)\ \Sigma_{ij} [\, A_{ij} - k_i k_j/2m \,]\ \delta(c_i, c_j)$$

- Modularity maximization over the weighted account graph.
- No fixed cluster count is imposed — Louvain finds it.
- Applied to 668 accounts with ≥1 shared resource → 70 candidate clusters.

3 7 CHAMPION FEATURES

1	cluster_size	N (accounts in the cluster)
2	entity_reuse_ratio	1 − (distinct_resources / total_usages)
3	internal_density	2E / [N(N−1)]
4	referral_cycle_ratio	(nodes in referral cycles of size ≥2) / N
5	referral_overlap_ratio	internal_referral_edges / total_referral_edges_touching_cluster
6	median_activation_days	median(first_order_date − referral_date), over referred accounts
7	referral_density	internal_referral_pairs / [N(N−1)/2]

- 3 structural (resource-sharing) features + 4 directed-referral-graph features.
- Referral rings deliberately avoid shared resources to evade the structural features alone — these 4 catch them via cycle structure and activation timing instead.

4 CALIBRATED CLASSIFIER

$$p(\text{fraud}) = \sigma(\, \beta_0 + \Sigma\, \beta_i \cdot z_i \,),\ z_i = (x_i - \mu_i)/\sigma_i$$
$$\sigma(x) = 1 / (1 + e^{-x})$$

- StandardScaler + Logistic Regression on the 7 champion features.
- 5-fold Stratified Cross-Validation produces out-of-fold probabilities — every cluster is scored by a model that never saw it during training.

5 COST-OPTIMAL THRESHOLD

$$\text{FP\_Cost}(t) = \text{fp\_multiplier} \times \text{avg\_order\_value} \times \Sigma(\text{non\_ring\_accounts\_flagged})$$
$$\text{FN\_Cost}(t) = \Sigma(\text{ring\_value\_missed})$$
$$\text{Total\_Cost}(t) = \text{FP\_Cost}(t) + \text{FN\_Cost}(t)$$

- Every candidate probability is tested as threshold t, cost pooled across 15 synthetic seeds (954 clusters, 292 true rings).
- Locked threshold = plateau midpoint = 0.1333 — minimizes pooled cost at Rs 1,11,282, avg order value ≈ Rs 890.

6 VALIDATION: BOOTSTRAP + TEMPORAL BACKTEST

Percentile Bootstrap

$$CI = [\ P2.5(\theta^*),\ P97.5(\theta^*)\ ],$$
B = 10,000 resamples of 70 clusters, with replacement

- Precision CI: [1.000, 1.000]
- Recall CI: [1.000, 1.000]

Zero-Lookahead Reconstruction

G_T = build_graph(all_records WHERE timestamp ≤ T)
Repeated at 101 snapshots, 7-day cadence

- 28/28 rings detected
- Median Latency 30.0 days
- 90.4% fraud volume intercepted pre-clearance
- Rs 6.71L counterfactual value protected

Bootstrap answers 'how much would a different sample have changed the answer.' The temporal engine answers a different question: rebuilding the graph as of each past date, using only information available then, to rule out lookahead leakage.